

A 21-day Daniel fast with or without krill oil supplementation improves anthropometric parameters and the cardiometabolic profile in men and women

Background The Daniel Fast is a vegan diet that prohibits the consumption of animal products, refined foods, white flour, preservatives, additives, sweeteners, flavorings, caffeine, and alcohol. Following this dietary plan for 21 days has been demonstrated to improve blood pressure, LDL-C, and certain markers of oxidative stress, but it has also been shown to lower HDL-C. Krill oil supplementation has been shown to increase HDL-C. **Methods** We investigated the effects of following a Daniel Fast dietary plan with either krill oil supplementation (2 g/day) or placebo supplementation (coconut oil; 2 g/day) for 21 days. The subjects in this study (12 men and 27 women) were heterogeneous with respect to body mass index (BMI) (normal weight, overweight, and obese), blood lipids (normolipidemic and hyperlipidemic), blood glucose (normal fasting glucose, impaired fasting glucose, and type 2 diabetic), and blood pressure (normotensive and hypertensive). **Results** Krill oil supplementation had no effect on any outcome measure (all $p > 0.05$), and so the data from the krill oil group and the placebo group were collapsed and analyzed to examine the effects of following a 21-day Daniel Fast. Significant reductions were observed in LDL-C (100.6 ± 4.3 mg/dL vs. 80.0 ± 3.7 mg/dL), the LDL:HDL ratio (2.0 ± 0.1 vs. 1.7 ± 0.1), fasting blood glucose (101.4 ± 7.5 mg/dL vs. 91.7 ± 3.4 mg/dL), fasting blood insulin (7.92 ± 0.80 μ U/mL vs. 5.76 ± 0.59 μ U/mL), homeostasis model assessment of insulin resistance (HOMA-IR) (2.06 ± 0.30 vs. 1.40 ± 0.21), systolic BP (110.7 ± 2.2 mm Hg vs. 105.5 ± 1.7 mm Hg), and body weight (74.1 ± 2.4 kg vs. 71.5 ± 2.3 kg) (all $p < 0.05$). **Conclusion** Following a Daniel Fast dietary plan improves a variety of cardiometabolic parameters in a wide range of individuals in as little as 21 days, and these improvements are unaffected by krill oil supplementation. Trial registration Clinicaltrial.govNCT01378767